

CHAPTER 5.2B: DIFFERENTIAL PRIVACY

FOUNDATIONS OF DISTRIBUTED SYSTEMS

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SETTING & GOALS

- We have a database D with sensitive information about individuals
- There is a **trusted** database curator with full access to the database
- **Data analysts** can ask queries to the curator, and the curator responds to each query S with the response R_S .
- **The analysts can be malicious.**
- The curator uses a **mechanism** M to determine the query response, i.e.,
 $M : S \rightarrow R_S$
- We want the mechanism to satisfy two properties:
 1. Give statistically useful responses to the analysts (*utility*)
 2. Without leaking sensitive information about the individuals in the database (*privacy*)

- Simple mechanism: always report the correct query result
 - Statistically useful!
 - But leaks information about individuals
 - Even if we limit query types!
 - Differencing attack: we can leak information about individuals by aggregating only over large groups
- What if we report the correct query result, **plus some noise**?
 - Adding noise is tricky!
 - Independent noise? → Attack with repeated queries, median filter
 - Too little noise? → Dinur-Nissim reconstruction attack

1. Dinur-Nissim reconstruction attack
2. Differential privacy
3. Properties of differential privacy
4. Mechanisms for differential privacy
5. Approximate differential privacy

1. DINUR-NISSIM RECONSTRUCTION ATTACK

- If the mechanism adds too little noise when responding to aggregated queries, an adversary can reconstruct the database with high accuracy and efficiency.
- Such mechanisms are called **blatantly non-private**.

ATTACK SETUP

- We have a database D with n records: $D = \{d_1, d_2, \dots, d_n\}$, where each record corresponds to one individual
- Each record d_i may consist of k attributes
- We assume the adversary knows $k - 1$ attributes for each record; the only unknown attribute is a single bit b_i

$$D = \begin{bmatrix} a_{1,1} & a_{1,2} & \dots & a_{1,k-1} & b_1 \\ a_{2,1} & a_{2,2} & \dots & a_{2,k-1} & b_2 \\ \vdots & \vdots & \dots & \vdots & \vdots \\ a_{n,1} & a_{n,2} & \dots & a_{n,k-1} & b_n \end{bmatrix}$$

- The attacker can ask aggregated queries, e.g., counting queries: how many records in D , satisfying a condition $C(a_{*,1}, \dots, a_{*,k-1})$ have their secret bit = 1?

ATTACK EXAMPLE

Name	ZIP	DOB	Covid
Alice	2345	May 2 1984	1
Bob	9490	Feb 8 2001	0
Charlie	4001	Oct 10 1960	1
David	5642	Apr 5 1977	0
Eve	3357	Jan 17 1987	1
...	...	...	...

- How many rows with condition (Name = “Alice” AND DOB > 1980) have Covid=1?
- The adversary can use background knowledge to formulate conditions and queries → can select arbitrary rows in the database

- A query S is a binary vector indicating which rows the adversary has selected:
 $S = [0, 1, 1, 0, \dots, 0]$
- The curator first calculates the true answer $A(S) = S \times [b_1, b_2, \dots, b_n]$
- Then the curator adds random noise E to the true answer and returns the value $R_S = A(S) + E$
- R_S is a single number: the result of a counting or other aggregate query

ATTACK #1 (EXPONENTIAL)¹

If the analyst is allowed to ask 2^n queries to a dataset of n users, and the curator adds noise with some bound E (linear in n), then based on the results, the adversary can reconstruct the database in all but at most $4E$ positions.

- 2^n queries means the adversary can issue every possible query
- The adversary then checks every possible candidate database C
- if $\exists S : |\sum_{i \in S} C[i] - R_S| > E$: rule out C (i.e., if there is a query for which the number of bits in which the candidate and the curator's noisy response differ is greater than E)
- All other candidate databases differ from the correct one by $4E$ at most
- For example, if $E = \frac{n}{400}$, adversary reconstructs 99% of entries!

¹I. Dinur and K. Nissim, “Revealing information while preserving privacy,” in *Proceedings of the Twenty-Second ACM SIGMOD-SIGACT-SIGART Symposium on Principles of Database Systems*, ser. PODS '03, San Diego, CA, USA: Association for Computing Machinery, Jun. 2003, pp. 202–210. doi: 10.1145/773153.773173. [Online]. Available: <https://doi.org/10.1145/773153.773173>.

ATTACK #2 (POLYNOMIAL)³

If the analyst is allowed to ask $O(n)$ queries to a dataset of n users, and the curator adds noise with some bound $E = o(\sqrt{n})$, then based on the results, a computationally efficient adversary can reconstruct the database in all but at most $o(n)$ positions.

A remark on notation: “Big-O means *is of the same order as*. The corresponding little-o means *is ultimately smaller than*” ²

²<https://www.stat.cmu.edu/~cshalizi/uADA/13/lectures/app-b.pdf>

³I. Dinur and K. Nissim, “**Revealing information while preserving privacy,**” in *Proceedings of the Twenty-Second ACM SIGMOD-SIGACT-SIGART Symposium on Principles of Database Systems*, ser. PODS '03, San Diego, CA, USA: Association for Computing Machinery, Jun. 2003, pp. 202–210. doi: 10.1145/773153.773173. [Online]. Available: <https://doi.org/10.1145/773153.773173>.

A mechanism is **blatantly non-private** if an adversary can reconstruct a database that matches with the true database in all but $o(n)$ entries.

- According to the polynomial attack, adding a noise of $o(\sqrt{n})$ is blatantly non-private.
- Consider the magnitude of this error in the context of the database size: in a database with 25 rows, adding noise with order 5 or less is blatantly non-private (is the data still useful?)
- This definition does not specify whether a mechanism is private.
- Instead, it defines a criteria to show that a mechanism is clearly not private.

2. DIFFERENTIAL PRIVACY

We follow the same setting as for the reconstruction attack:

- Trusted database D
- The curator runs an algorithm A over D and releases data T
- The adversary has access to T and knows the details of A

Privacy notion:

- It is not important whether the adversary can reconstruct the database
- But the adversary should learn (almost) **nothing new** about an individual after seeing A and T , regardless of what **other data** the adversary knows.

NO ASSUMPTIONS OR LIMITS ON THE ADVERSARY'S BACKGROUND KNOWLEDGE!

- If the adversary does not know whether I am in the database, it won't know that either after seeing the result
- If the adversary already knows whether I am in the database, it won't know more about the secret values I supplied.

EXAMPLE FROM THE ADVERSARY'S POINT OF VIEW

- You know about some very challenging university course and you want to know if Alice has completed the course
- Background knowledge 1: Alice is a top student who always scores at least 90%
- Background knowledge 2: The course is so hard that most students score around 50%
- Background knowledge 3: 30 students took the course this year
- The database curator allows the following access:
 - You can make 5 queries
 - For each query, the curator selects 3 random students and returns the average score

- The adversary observes the outputs of the 5 queries
- There are two possible cases: (1) Alice took the course, (2) Alice didn't take the course
- What are the query outputs in case (2)?

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- There are two possible cases: (1) Alice took the course, (2) Alice didn't take the course
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- We expect outputs very close to $[50, 50, 50, 50, 50]$
- What are the outputs in case (1)?

- The adversary observes the outputs of the 5 queries
- There are two possible cases: (1) Alice took the course, (2) Alice didn't take the course
- What are the query outputs in case (2)?
- We expect outputs very close to $[50, 50, 50, 50, 50]$
- What are the outputs in case (1)?
- Each query output has a 10% chance to include Alice⁴
- If Alice is included, we expect a response of 63 $((50 + 50 + 90)/3)$
- For 5 independent queries, the chance that at least one query includes Alice is $1 - (1 - 0.1)^5 = 0.4095$

⁴she is picked with probability $\frac{1}{30} \frac{29}{29} \frac{28}{28} + \frac{29}{30} \frac{1}{29} \frac{28}{28} + \frac{29}{30} \frac{28}{29} \frac{1}{28} = 0.1$

WHAT WENT WRONG?

- Alice's score has too much impact on the output!
- As a result, seeing the output of the algorithm allows the attacker to differentiate which database is the underlying database representing the class score.
- This is exactly what Differential Privacy tries to capture!
- Informally: any single element in a dataset should have only a limited impact on the output

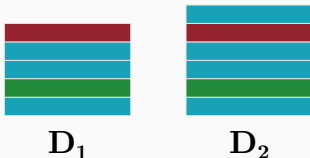
- A mechanism $M : X \rightarrow Y$ is ϵ -differentially private (ϵ -DP) if for any two neighboring databases D_1 and D_2 :

$$\forall T \subseteq Y : \Pr[M(D_1) \in T] \leq e^\epsilon \Pr[M(D_2) \in T]$$

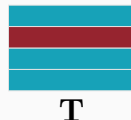
- Neighboring databases: two databases D_1 and D_2 are neighboring if they agree except for a single entry.
- All known mechanisms that satisfy differential privacy are **randomized functions**

⁵C. Dwork, “Differential privacy,” in *International Colloquium on Automata, Languages, and Programming (ICALP)*, ser. LNCS, vol. 4052, Venice, Italy: Springer, Jul. 2006, pp. 1–12. [Online]. Available: http://link.springer.com/chapter/10.1007/11787006_1.

For every pair of inputs
that differ in one row



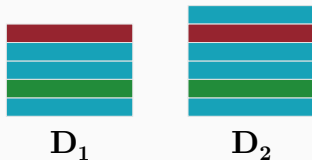
For every output



The adversary should not be able to distinguish between any D_1 and D_2 based on any T , i.e.,

$$\ln \left(\frac{\Pr[M(D_1) = T]}{\Pr[M(D_2) = T]} \right) \leq \epsilon, \epsilon > 0$$

For every pair of inputs
that differ in one row

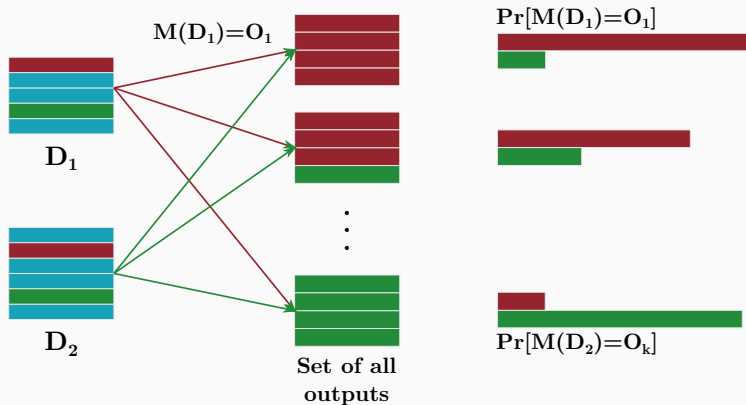


For every output



- Why pairs of datasets that **differ in one row**? → Simulate presence/absence of a single record
- Why **all pairs of datasets**? → Guarantee holds no matter what the other records are (arbitrary background knowledge!)

FORMAL DEFINITION, EXPLAINED



- Why **all outputs**? → Should not be able to distinguish whether input was D_1 or D_2 no matter what the output

PRIVACY PARAMETER ϵ (ALSO “PRIVACY BUDGET”)

- Controls the degree to which D_1 and D_2 can be distinguished
- Smaller ϵ means more privacy
- Remember: $Pr[M(D_1) \in T] \leq e^\epsilon Pr[M(D_2) \in T]$
- For small epsilon: $e^\epsilon \approx 1 + \epsilon$
- What is “small”⁶?
 - Think $\epsilon = 0.01$ or $\epsilon = 0.1$
 - But: choice of ϵ is a social question
 - If probability of harm is very small, $\epsilon = \ln 3 \approx 1.1$ may be acceptable

⁶C. Dwork, “Differential privacy: A survey of results,” in *Theory and Applications of Models of Computation*, Springer, 2008, pp. 1–19. [Online]. Available: http://link.springer.com/chapter/10.1007/978-3-540-79228-4_1.

3. PROPERTIES OF DIFFERENTIAL PRIVACY

- Suppose mechanism $M : X \rightarrow Y$ is ϵ -differentially private. Then, for any algorithm $A : Y \rightarrow Z$, we have that $A \circ M : X \rightarrow Z$ is also ϵ -differentially private.
- Processing the result of a differentially private mechanism with any arbitrary algorithm does not change the privacy guarantee!
- Once the data is privatized, it cannot be “un-privatized”

⁷C. Dwork and A. Roth, *The Algorithmic Foundations of Differential Privacy*. Now Publishers, Aug. 2014.

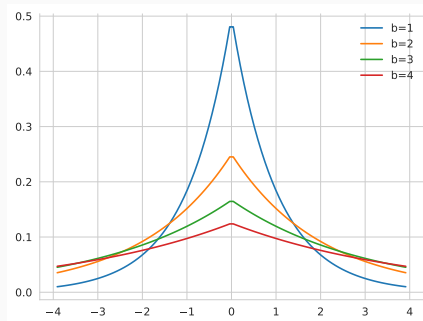
- If we apply two ϵ -differentially private mechanisms to data, the combined output is 2ϵ -differentially private
- Formally: given two mechanisms, $M_1 : X \rightarrow Y_1$ being ϵ_1 -DP, and $M_2 : X \rightarrow Y_2$ being ϵ_2 -DP.
- We define a new mechanism $M : X \rightarrow Y_1 \times Y_2$ as $M(X) = (M_1(X), M_2(X))$. Then M is $(\epsilon_1 + \epsilon_2)$ -DP.
- This also holds when we apply the same mechanism repeatedly (repeated queries!)

- Suppose mechanism $M : X \rightarrow Y$ is ϵ -differentially private.
- Suppose we split our dataset X into k disjoint chunks, i.e. $X = x_1 \cup x_2 \cdots \cup x_k$
- Then the mechanism which releases all of the results $M(x_1), \dots, M(x_k)$ satisfies ϵ -differential privacy
- Why? We apply the same mechanism k times, so by sequential composition, it should satisfy $k\epsilon$ -DP
- But: if each individual contributes only one row to X , then each individual contributes to only *one* of the chunks $x_1, \dots, x_k$
- This means that the mechanism M sees each individual's data only once, so the privacy cost ϵ is appropriate for all individuals in X .

- Suppose mechanism $M : X \rightarrow Y$ is ϵ -differentially private.
- Suppose D_1 and D_2 are two datasets which differ in exactly k positions. Then:
- $\forall T \subseteq Y : Pr[M(D_1) \in T] \leq e^{k\epsilon} Pr[M(D_2) \in T]$
- If you need to hide the “effect” of a whole group, you need to prepare a larger privacy budget.

4. MECHANISMS FOR DIFFERENTIAL PRIVACY

- Idea: the curator computes the correct result $A(S)$, then samples noise η from the Laplace distribution and returns $A(S) + \eta$
- The probability density function for $\text{Lap}(\mu, b)$ is: $f(x) = \frac{1}{2b} \exp\left(\frac{-|x-\mu|}{b}\right)$
- For DP, usually $\mu = 0$
- So $\text{Lap}(b)$ means $\text{Lap}(0, b)$
- Variance of $\text{Lap}(b)$: $\sigma^2 = 2b^2$
- How do we choose parameter b ?



- We want to choose noise so that:
 - Each answer does not leak too much information about the database.
 - Noisy answers are close to the original answers.
 - $\rightarrow$ the noise level needs to depend somehow on the magnitude of the possible answers!
- Consider a query $q : X \rightarrow R$.
- **Sensitivity** $S(q)$ is the smallest number so that for any neighboring tables D_1, D_2 , $|q(D_1) - q(D_2)| \leq S(q)$
- In other words, sensitivity corresponds to the largest difference that a single entry can make in query output

- The ℓ_1 -sensitivity of a real-valued query function q is

$$\Delta_1^q = \max_{D_1, D_2: d(D_1, D_2) \leq 1} |q(D_1) - q(D_2)|$$

- Differential privacy for the Laplace mechanism: **If the sensitivity Δ_1^q of query q is S , then ϵ -differential privacy is guaranteed if $b = \frac{S}{\epsilon}$**

- How many people have Covid?
- Query $q(x) = \sum_i x_i$
- Mechanism $M(x) = q(x) + Y$, where $Y \sim \text{Lap}(0, b)$

Covid (Y/N)

Y

N

Y

Y

N

N

- How many people have Covid?
- Query $q(x) = \sum_i x_i$
- Mechanism $M(x) = q(x) + Y$, where $Y \sim \text{Lap}(0, b)$
- Sensitivity = 1
- Parameter $b = \frac{S}{\epsilon} = \frac{1}{\epsilon}$
- So $Y \sim \frac{\epsilon}{2} e^{-\epsilon|y|}$

Covid (Y/N)
Y
N
Y
Y
N
N

- What is the total amount an organization pays out in salaries?
- What is the largest influence any single salary can have on the result?

Salary

2

7

3

8

11

4

- What is the total amount an organization pays out in salaries?
- What is the largest influence any single salary can have on the result?
- → We need to know the **range** of possible salaries!
- Here: sensitivity = largest salary = 11
- In general: sensitivity is unbounded if we do not know the range.
- $b = \frac{11}{\epsilon}$, so $Y \sim \frac{\epsilon}{22} e^{-\frac{\epsilon|y|}{11}}$

Salary
2
7
3
8
11
4

- What is the average score of students in a given course?
- We know that scores are $\in [0, 100]$
- Is this enough to compute sensitivity?

Score

24

78

31

86

11

...

46

SENSITIVITY: AVERAGE QUERY

- What is the average score of students in a given course?

- We know that scores are $\in [0, 100]$

- Is this enough to compute sensitivity?

- No! We also need to know the number of students

$$\begin{aligned}\ell_1 &= \max(|\frac{k_1 + \sum_{29 \text{ students}}}{30} - \frac{k_2 + \sum_{29 \text{ students}}}{30}|) \\ &= \frac{1}{30} \max(|k_1 - k_2|) \quad ^8 \\ &= \frac{100}{30}\end{aligned}$$

- $b = \frac{10}{3\epsilon}$, so $Y \sim \frac{3\epsilon}{20} e^{-\frac{3\epsilon|y|}{10}}$

Score

24

78

31

86

11

...

46

⁸The maximum is when $k_1 = 0$ and $k_2 = 100$ or vice versa.

- We can clip the values in our dataset to enforce the bounds for average or summation queries
- But: we **cannot** determine the bounds by looking at the data because **our mechanism would not satisfy differential privacy anymore!**
 - If we need to do so, we need to use differentially private queries, and include the epsilons in the total privacy cost of our algorithm
 - For example: if we use a sequence of DP counts with $\epsilon = 0.01$ to determine bounds on the *Age* column, and we aim for $\epsilon = 1$ for a summation query, we need to use $\epsilon = 0.99$ for the sum

- The Laplace mechanism works for any function that returns a real number
- For utility, we consider the expected error introduced by the noisy answer, i.e.,

$$\begin{aligned} & E(\text{true answer} - \text{noisy answer})^2 \\ &= \text{Var}(\text{Lap}(S(q)/\epsilon)) \\ &= 2 \frac{S(q)^2}{\epsilon^2} \end{aligned}$$

- The error increases with larger sensitivity and smaller ϵ
- To get more useful answers, we should aim to design queries that minimize sensitivity!

- Many interesting queries do not return a real number
 - What is the most common nationality?
 - What is the best day for a meeting?
- Perturbation can make outputs invalid
 - Outputs may need to be integer values
 - Outputs may need to be non-negative
- In general: we want to select the **best** element from a set while preserving differential privacy

⁹F. McSherry and K. Talwar, “**Mechanism Design via Differential Privacy**,” in *48th Annual IEEE Symposium on Foundations of Computer Science (FOCS’07)*, Oct. 2007, pp. 94–103. DOI: 10.1109/FOCS.2007.66.

SCORING FUNCTION

- We have a dataset D
- We define the set of outputs R from which we want to select the best element
- We define a **scoring function** $u : D \times R \rightarrow \mathbb{R}$ with sensitivity Δu
- We are interested in the element with the highest score
- The exponential mechanism returns an element which may (or may not) have the highest score

Name	Country
Alice	USA
Bob	Germany
Charlie	USA
David	Switzerland
Eve	Austria
...	...

Set of outputs $R = \{\text{Austria, Germany, Switzerland, USA}\}$

$$u(D, R) = \#(D, R)$$

- Sensitivity of the scoring function is defined very similarly to sensitivity of queries

$$\Delta u = \max_{R, D, D'} |u(D, R) - u(D', R)|$$

- The only change is that the maximum is taken over all R , in addition to all D, D' that differ in one entry
- What is Δu in the nationalities example?

- Given input d , set of outputs R , and scoring function u , the exponential mechanism samples an output $r \in R$ with probability proportional to

$$\exp\left(\frac{\epsilon u(d, r)}{2\Delta u}\right)$$

- Can we realize the question “What is the most common nationality?” with the Laplace mechanism?

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- Yes!
 1. For each nationality r , calculate its noisy score: $u(d, r) + \text{Lap}(\frac{\Delta u}{\epsilon})$
 2. Output the noisy scores, the analyst picks the maximum noisy score
- What level of privacy does this algorithm satisfy?

- Can we realize the question “What is the most common nationality?” with the Laplace mechanism?
- Yes!
 1. For each nationality r , calculate its noisy score: $u(d, r) + \text{Lap}(\frac{\Delta u}{\epsilon})$
 2. Output the noisy scores, the analyst picks the maximum noisy score
- What level of privacy does this algorithm satisfy?
 - We apply step (1) $|R|$ times. The sensitivity of u is 1
 - By sequential composition, our algorithm satisfies $|R|\epsilon$ -differential privacy
- Why is the exponential mechanism (which satisfies ϵ -DP) so much better?
 - It releases **less information**: only the identity of the element with maximum noisy score, but not the score, and not the scores of the other elements!
 - If we modify step 2: *Output the element with the maximum noisy score*, then the algorithm becomes ϵ -differentially private¹⁰

¹⁰C. Dwork and A. Roth, *The Algorithmic Foundations of Differential Privacy*. Now Publishers, Aug. 2014.

5. APPROXIMATE DIFFERENTIAL PRIVACY

- A mechanism $M : X \rightarrow Y$ is (ϵ, δ) -differentially private $((\epsilon, \delta)$ -DP) if for any two neighboring databases $D_1 : X$ and $D_2 : X$:

$$\forall T \subseteq Y, \Pr[M(D_1) \in T] \leq e^\epsilon \Pr[M(D_2) \in T] + \delta$$

- Meaning of ϵ : limit the difference in output for neighboring databases which results in a privacy guarantee
- Meaning of δ : introduce a “failure probability” for privacy
 - With probability $1 - \delta$: same guarantee as ϵ -DP
 - With probability δ : no privacy guarantee at all
- Why? We need to add much less noise in the approximate-DP model!

¹¹C. Dwork, K. Kenthapadi, F. McSherry, *et al.*, “Our data, ourselves: Privacy via distributed noise generation,” in *Advances in Cryptology-EUROCRYPT 2006*, Springer, 2006, pp. 486–503. [Online]. Available: http://link.springer.com/chapter/10.1007/11761679_29.

- With probability δ , we lose the privacy guarantee
- In the worst case, the entire dataset could be released!
- We need δ to be very small
- Typical choice: less than $\frac{1}{n^2}$, where n is the size of the dataset
- In practice: mechanisms for approximate differential privacy do not fail catastrophically

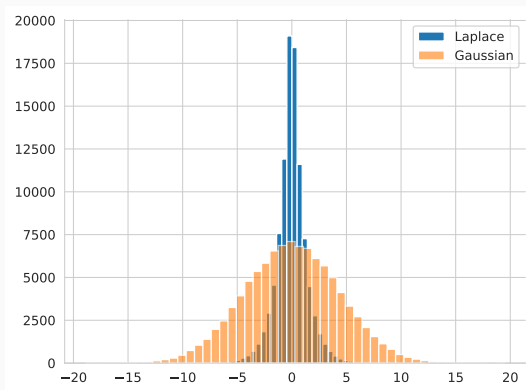
- Similar properties as ϵ -differential privacy
- Sequential composition
 - If $F_1(x)$ satisfies (ϵ_1, δ_1) -differential privacy
 - And $F_2(x)$ satisfies (ϵ_2, δ_2) -differential privacy
 - Then $G(x) = (F_1(x), F_2(x))$ satisfies $(\epsilon_1 + \epsilon_2, \delta_1 + \delta_2)$ -differential privacy
- Post-processing
- Parallel composition

- For a query function $f(x)$, the Gaussian mechanism returns $G(x) = f(x) + \mathcal{N}(\sigma^2)$
- The Gaussian mechanism satisfies (ϵ, δ) -differential privacy if $\sigma^2 = \frac{2s^2 \ln(1.25/\delta)}{\epsilon^2}$
- where s is the sensitivity of $f(x)$ and $\mathcal{N}(\sigma^2)$ denotes sampling from the Gaussian distribution with mean 0 and variance σ^2

¹²C. Dwork and A. Roth, *The Algorithmic Foundations of Differential Privacy*. Now Publishers, Aug. 2014, p. 265.

LAPLACE VS GAUSSIAN NOISE

Compare the noise distributions for Laplace noise with $\epsilon = 1.0$ and Gaussian noise with $\epsilon = 1.0, \delta = 10^{-5}$:



The Gaussian mechanism seems to result in much more noise than the Laplace mechanism! What is it useful for?

- Query functions so far: real-valued, i.e., they return a single number:
 $f : D \rightarrow \mathbb{R}$
- Vector-valued functions return a vector of numbers: $f : D \rightarrow \mathbb{R}^k$
- Examples for vector-valued functions?

- Query functions so far: real-valued, i.e., they return a single number:
 $f : D \rightarrow \mathbb{R}$
- Vector-valued functions return a vector of numbers: $f : D \rightarrow \mathbb{R}^k$
- Examples for vector-valued functions?
 - Histograms: vector of bin counts
 - Sequences of queries, e.g., sums over different columns
 - Machine learning: feature vectors and gradients (more about this in the spring lecture “Privacy-Preserving Methods for Data Science and Distributed Systems”)
- Both Laplace and Gaussian mechanisms can be extended to vector-valued functions

- How did we define the sensitivity Δ_1^q of a real-valued query q ?

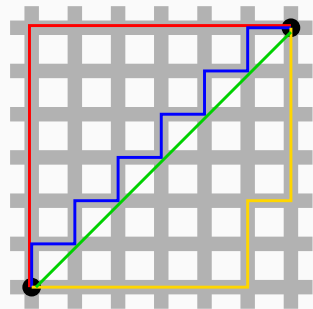
- How did we define the sensitivity Δ_1^q of a real-valued query q ?
 - We used the maximum **magnitude** of the difference between the query function applied to neighboring datasets:

$$\Delta_1^q = \max_{D_1, D_2: d(D_1, D_2) \leq 1} |q(D_1) - q(D_2)|$$

- For vector-valued functions: there are different ways to compute the magnitude of a vector!

SENSITIVITIES OF VECTOR-VALUED FUNCTIONS

- L1 norm:
 - The sum of the vector's elements
 - $\|V\|_1 = \sum_{i=1}^k |V_i|$
 - In 2-d: Manhattan or Taxicab distance (red, yellow, and blue lines)
- L2 norm:
 - The square root of the sum of the squares
 - $\|V\|_2 = \sqrt{\sum_{i=1}^k V_i^2}$
 - In 2-d: Euclidean distance (green line)
- The sensitivity of vector-valued function relies on these norms:
 - L1 sensitivity $\Delta_1^q = \max_{d(D_1, D_2) \leq 1} \|q(D_1) - q(D_2)\|_1$
 - L2 sensitivity $\Delta_2^q = \max_{d(D_1, D_2) \leq 1} \|q(D_1) - q(D_2)\|_2$



Source:

https://upload.wikimedia.org/wikipedia/commons/0/08/Manhattan_distance.svg

MECHANISMS FOR VECTOR-VALUED FUNCTIONS

- L2 sensitivity is generally smaller (for large vectors *much* smaller) than L1 sensitivity
- Remember: small sensitivity gives us better accuracy and/or more privacy!
- Unfortunately, we are **not** free to choose which sensitivity we use.
- The vector-valued Laplace mechanism releases $f(x) + Y_1, \dots, Y_k$, where Y_i are drawn i.i.d. from the Laplace distribution with scale parameter $b = \frac{s}{\epsilon}$ and s is the **L1** sensitivity of f
- The vector-valued Gaussian mechanism releases $f(x) + Y_1, \dots, Y_k$, where Y_i are drawn i.i.d. from the Gaussian distribution with $\sigma^2 = \frac{2s^2 \ln(1.25/\delta)}{\epsilon^2}$ and s is the **L2** sensitivity of f

MECHANISMS FOR VECTOR-VALUED FUNCTIONS

- L2 sensitivity is generally smaller (for large vectors *much* smaller) than L1 sensitivity
- Remember: small sensitivity gives us better accuracy and/or more privacy!
- Unfortunately, we are **not** free to choose which sensitivity we use.
- The vector-valued Laplace mechanism releases $f(x) + Y_1, \dots, Y_k$, where Y_i are drawn i.i.d. from the Laplace distribution with scale parameter $b = \frac{s}{\epsilon}$ and s is the L1 sensitivity of f

This is why the Gaussian mechanism is useful!

Gaussian mechanism releases $f(x) + Y_1, \dots, Y_k$, where Y_i are drawn i.i.d. from the Gaussian distribution with $\sigma^2 = \frac{2s^2 \ln(1.25/\delta)}{\epsilon^2}$ and s is the L2 sensitivity of f

1. Dinur-Nissim reconstruction attack
2. Differential privacy
3. Properties of differential privacy
4. Mechanisms for differential privacy
5. Approximate differential privacy